

Part I: Multiple Choice

1. How many squares does a four-variable K-map contain, and why? [2 pts]
 - ☐ A. 8, because there are 8 possible product terms.
 - ☐ B. 16, because four variables produce 2^4 distinct minterms, one per square.
 - ☐ C. 12, because each variable contributes three squares.
 - ☐ D. 4, one square for each variable.
2. When two adjacent squares are grouped, the resulting product term has: [2 pts]
 - ☐ A. one more literal than a single square.
 - ☐ B. exactly one literal, no matter how large the map.
 - ☐ C. one fewer literal, because the variable that changes between the squares drops out.
 - ☐ D. the same number of literals as a single square.
3. To obtain a minimized *product-of-sums* expression from a K-map, you should: [2 pts]
 - ☐ A. group the 1's and OR the resulting product terms.
 - ☐ B. group the 0's, write a sum term for each group using the complemented reading rule, then AND the sum terms.
 - ☐ C. group the 1's and complement the final expression.
 - ☐ D. group only the don't-cares.
4. When a don't-care minterm (X) appears on a K-map, you should: [2 pts]
 - ☐ A. always include it in a group, since every X must be covered.
 - ☐ B. never include it in any group.
 - ☐ C. treat it as a 1 only when doing so enlarges a group, and otherwise leave it uncovered.
 - ☐ D. treat it as a 0 that must be grouped with the other 0's.
5. The exclusive-OR output $x \oplus y$ equals 1 exactly when: [2 pts]
 - ☐ A. both inputs are 1.
 - ☐ B. both inputs are 0.
 - ☐ C. the two inputs have different values.
 - ☐ D. at least one input is 1.
6. A multi-input XOR outputs 1 when: [2 pts]
 - ☐ A. an odd number of its inputs are 1.
 - ☐ B. an even number of its inputs are 1.
 - ☐ C. all of its inputs are 1.
 - ☐ D. none of its inputs are 1.

Part II: Show Work

The K-maps printed in this exam are **blank grids only**. For each one you must: (1) write the **correct variable order** on the diagonal lines (2) label each row and column with the correct **input combination** (3) be able to identify **which square corresponds to which minterm**. You may lightly write the minterm number m_i in the corner of each square if it helps you. Then place your 1's, 0's, or X's, circle your groups, and read off the requested expression. Partial credit will be given.

1. Truth table to three-variable K-map. Consider

$$F(a, b, c) = a'c + ab' + bc.$$

- (a) Complete the truth table below. Fill in each intermediate column first, then combine them to get F . [6 pts]

a	b	c	$a'c$	ab'	bc	F
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

- (b) Transfer the minterms to the blank K-map (label rows, columns, and combinations yourself), then write the minimal sum-of-products expression. [8 pts]

Minimal SOP: _____

2. Four-variable K-map (sum of products). Simplify

$$F(w, x, y, z) = \sum m(3, 4, 5, 6, 7, 11, 15)$$

to a minimal sum-of-products expression. Fill in the blank K-map (label everything yourself), circle your groups, and write the result. [12 pts]

Minimal SOP: _____

3. **Don't-care conditions.** Simplify the incompletely specified function

$$F(a, b, c, d) = \sum m(0, 2, 3, 9, 13), \quad d(a, b, c, d) = (1, 11, 15)$$

to a minimal sum-of-products expression. Mark and simplify the K-map, then write your result on the line below. **[12 pts]**

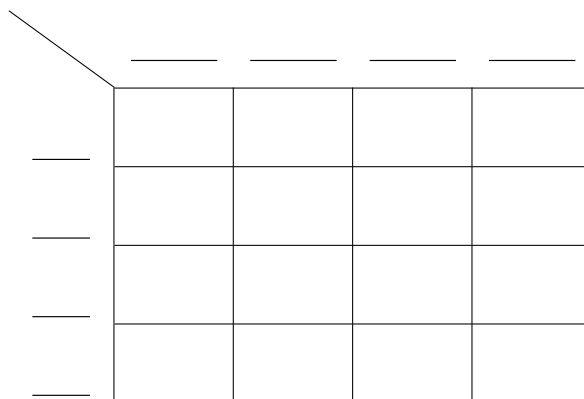
Minimal SOP: _____

4. **BCD don't-care problem.** A circuit reads a binary-coded-decimal digit $abcd$ (valid inputs 0 through 9 only) and outputs $F = 1$ exactly when the digit is **even** (i.e. 0, 2, 4, 6, 8). The combinations 10 through 15 never occur and are don't-cares:

$$F(a, b, c, d) = \sum m(0, 2, 4, 6, 8), \quad d(a, b, c, d) = (10, 11, 12, 13, 14, 15).$$

Find the minimal sum-of-products expression using the blank K-map.

[10 pts]

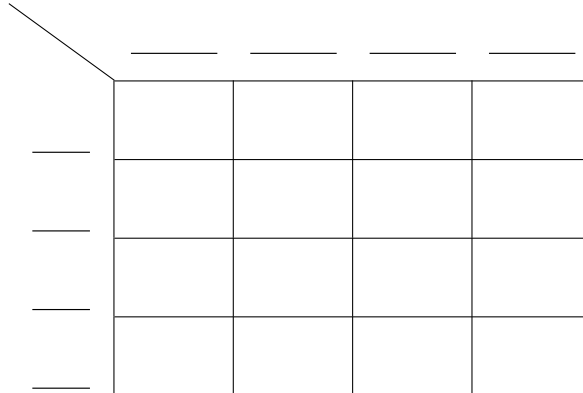


Minimal SOP: _____

5. **NAND implementation (two faces of NAND).** Simplify

$$F(w, x, y, z) = \sum m(2, 3, 6, 7, 8, 9, 10, 11, 15)$$

with a four-variable K-map, then draw the **NAND** circuit for your simplified expression using the two-faces-of-NAND method shown in class. [12 pts]

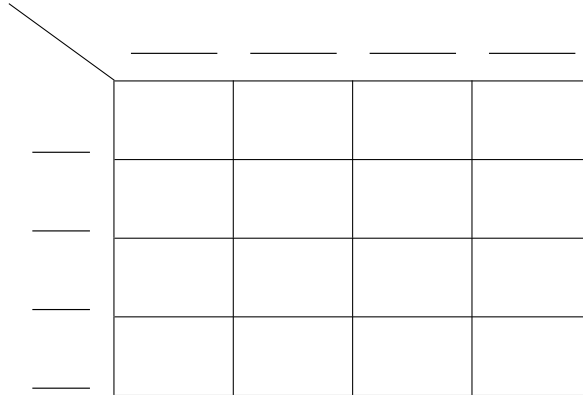


Minimal SOP: _____

6. **NOR implementation (two faces of NOR).** Simplify

$$F(a, b, c, d) = \prod M(0, 1, 2, 3, 4, 8, 12)$$

with a four-variable K-map, then draw the **NOR** circuit for your simplified expression using the two-faces-of-NOR method shown in class. [12 pts]



Minimal POS: _____

7. XOR / XNOR and equivalence. Answer each part.

- (a) Simplify to a minimum expression, showing the identities you use:

$$a \oplus 0 \oplus a \oplus b.$$

[5 pts]

- (b) Express the odd function $A \oplus B \oplus C$ as a sum of minterms $\sum m(\dots)$, and explain in one sentence why exactly those minterms appear. **[5 pts]**

- (c) We want to check whether two 4-bit numbers $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$ are equal. Write the Boolean expression for the output **Equal**. Then determine whether $A = 1011$ and $B = 1001$ are equal, showing the bit-by-bit comparison. [6 pts]

- (d) A 4-bit message $wxyz = 1101$ is sent using **even** parity. Find the parity bit P and write the full transmitted word $wxyzP$. [4 pts]